Trust-Aware Context Selection: Learned Signal Combination for Multi-Hop Evidence Retrieval

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Abstract

Retrieval-augmented language models increasingly depend on a pool of heterogeneous candidate passages, and the correctness of a grounded answer hinges on which passages enter the context window. Most systems collapse this selection to a single relevance score, yet a classical view of trust decomposes into several orthogonal dimensions: relevance to the query, intrinsic authority of the source, and corroboration across independent sources. We ask whether combining six lightweight signals covering these three dimensions yields better selection than any single signal, and whether the textbook intuition for combining them is empirically correct. On HotpotQA's distractor setting, with exactly two gold paragraphs among ten candidates per question, we compare a naive uniform average of three trust signals against a six-feature logistic regression fit on only one hundred questions per seed. The learned combiner attains Precision@2 of 0.345 ± 0.032, a 3.3 absolute-point improvement over the strongest single-signal baseline SBERT at 0.312 ± 0.035, with the gap consistent on every seed; the gold-versus-distractor score gap more than doubles, from 0.170 to 0.349, indicating substantially better calibration. The uniform average instead collapses to 0.080, below BM25. Inspecting the fitted weights diagnoses why: paragraph length and cross-paragraph consistency are learned as anti-signals, because distractors are length-matched to gold and topically close enough that cross-source similarity peaks among distractors rather than among the gold pair. In distractor-rich retrieval, the classical trust heuristics are correct in prior but wrong in sign, and supervised calibration over a handful of examples is sufficient to recover them.

1 Introduction

Large language models increasingly delegate factual recall to an external context: a set of retrieved documents, a cached memory fragment, a tool-returned snippet, or a user-supplied note. The correctness of a downstream answer therefore depends less on what the model itself knows and more on which evidence the pipeline chooses to place in its context window. In tasks such as medical question answering or legal reasoning, where evidence quality matters as much as topical match, that choice has to consider more than just relevance. A classical view of trust decomposes into orthogonal dimensions: *relevance* (does the source address the query?), *authority* (is the source intrinsically credible?), and *consistency* (do independent sources agree?). Two further dimensions, *provenance* (which specific source is this?) and *recency* (has the fact since moved?), require per-source metadata that many standard retrieval benchmarks, including the one we study, do not carry; we return to those two in the future-work discussion. We refer to the three measurable dimensions collectively as *trust signals*.

Most retrieval-augmented systems, however, collapse trust to a single number: a relevance score from a dense retriever or a sparse ranker. This is convenient, but the single-signal view is known to fail in predictable ways. Irrelevant-yet-topical distractors are scored high [1]; the position of the correct passage matters almost as much as its content [2]; and conflicting passages are merged without explicit adjudication [3]. A natural response is to compose a richer score from multiple signals. The intuition is simple: trustworthy equals relevant plus authoritative plus corroborated. Yet this intuition rests on an assumption that has not been tested empirically: that each individual trust heuristic actually points toward ground truth in the retrieval regimes we care about. In distractor-rich settings, where candidates are deliberately topically close, two textbook heuristics can in fact reverse sign.

We test this assumption on HotpotQA's distractor setting [4]. Each question is paired with ten candidate paragraphs (two gold, eight distractors) and sentence-level supporting-fact labels, so a paragraph-level scorer can be evaluated directly against a provenance oracle. We define six per-paragraph signals grouped into three trust dimensions: relevance (SBERT, BM25, TF-IDF), authority (log-length, question-entity overlap), and consistency (mean cross-paragraph SBERT cosine). We then contrast two combinations. The first averages three signals uniformly, following the textbook intuition above. The second fits a six-feature logistic regression on only one hundred questions per seed,

reserving two hundred for evaluation.

The learned combination lifts Precision@2 to **0.345 ± 0.032** versus SBERT's 0.312 ± 0.035, a 3.3 absolute-point improvement that holds on every seed. More tellingly, the calibration gap between gold and distractor scores more than doubles, from 0.170 to 0.349. The uniform combination, despite encoding the same textbook intuition, collapses to 0.080, well below every informed baseline including BM25 at 0.187. Inspecting the fitted weights explains both outcomes at once: in a distractor-rich pool, paragraph length and cross-paragraph consistency are learned as *anti*-signals, because distractors are both length-matched to gold and topically close enough to inflate cross-source similarity above that of the true gold pair. The classical trust heuristic is correct in prior but wrong in sign.

We make three contributions:

- We formalize paragraph-level trust-aware context selection as a per-candidate scoring problem over six orthogonal signals covering relevance, authority, and consistency, and release the signals as a drop-in CPU-runnable feature bank.

- We show that a six-feature logistic regression trained on one hundred supervised questions per seed beats the strongest single-signal baseline by 3.3 Precision@2 points and more than doubles the gold-versus-distractor score gap, with consistent sign across all three seeds.

- We identify a previously-undiagnosed failure mode of naive trust-signal averaging: in distractor-rich retrieval, two of the three signals that textbook intuition would combine with positive weight are learned with *negative* weight. The ablation isolates the phenomenon; the learned classifier recovers from it.

The rest of the paper is organized as follows. Section 2 surveys retrieval-augmented generation, multi-hop evidence selection, and source attribution. Section 3 formalizes the scoring problem and the two TrustScore variants. Section 4 describes the protocol on HotpotQA distractor. Section 5 reports the main comparison, the calibration diagnostic, and the learned weight pattern. Section 6 discusses scope, limitations, and a natural next step toward generation-time trust-aware selection.

2 Related Work

2.1 Multi-hop QA and supporting-fact selection

Multi-hop question answering evaluates the ability to combine evidence across multiple documents. HotpotQA [4] pairs each question with ten Wikipedia paragraphs and sentence-level supporting- fact annotations, making paragraph-selection quality directly measurable. A broader family of multi-hop benchmarks tests related phenomena: WikiHop [5] probes cross-document reasoning chains, 2WikiMultiHopQA [6] diagnoses reasoning-step coverage, IIRC [7] surfaces incomplete-context retrieval, and MuSiQue [8] composes single-hop questions into provably multi-hop ones to stress-test shortcut learning. Compositional-gap studies [9] and modular decompositions [10] further isolate where multi-hop models fail. A line of work frames supporting selection as a separate stage: cognitive graph retrievers [11] and iterative pathway retrievers [12] traverse linked entities across paragraphs, while transformer-based sentence selectors [13] predict supporting sentences directly. These methods are accurate but comparatively heavy. Our TrustScore sits strictly upstream of any such selector: it is pointwise, scores paragraphs in isolation, and does not reason over the candidate graph.

2.2 Retrievers and RAG benchmarks

Dense passage retrieval [14] replaced classical sparse retrievers on open-domain QA, and downstream variants including Fusion-in-Decoder [15], ColBERT [16, 17], and GTR-style generalisable dual encoders [18] pushed the retrieval-side recall ceiling further. More recent contrastive sentence-embedding recipes [19] underpin many zero-shot IR systems. Retrieval-quality benchmarking at scale is served by BEIR [20] for zero-shot IR and by the RGB benchmark [21] for RAG under noisy, counterfactual, and conflicting evidence. Classical fact-verification datasets FEVER [22] and FEVEROUS [23] motivated many of the evidence-selection metrics we borrow here.

2.3 Retrieval-augmented generation and context scoring

Retrieval-augmented generation [24, 25] retrieves a small context set at inference time to ground a generator, and a growing family of systems explores different retrieval-generation couplings: RETRO [26] attends to retrieved chunks during pre-training, Atlas [27] jointly fine-tunes retriever and generator, Active-RAG [28] retrieves selectively during decoding, REPLUG [29] treats the generator as a black box, Chain-of-Note [30] injects per-passage notes to improve robustness, and RECOMP [31] compresses retrieved context for bandwidth-constrained generators. Query-expansion tricks from large language models [32] shift some of this burden upstream. Recent work has probed the failure modes of single-signal retrieval: LLMs are sensitive to the position of the relevant context [2], are easily distracted by irrelevant context [33], struggle with on-topic distractors [1], require explicit context-faithful prompting

to avoid prior-knowledge shortcuts [34], and degrade further under conflicting passages [3]. Our work complements this line: rather than train a retriever end-to-end, we ask whether a *post-retrieval* trust-aware scorer, built from cheap signals and a small learned combiner, can separate gold from distractor candidates more reliably than a single relevance signal.

2.4 Source attribution and trust in LLM outputs

A parallel thread studies how language-model outputs should credit the evidence they stand on. GopherCite [35] and Attributed QA [36] fine-tune generators to produce quotation-backed answers; Gao et al. [37] survey attribution in generative systems and their RARR [38] post-hoc revises model outputs against retrieved evidence; Rashkin et al. [39] define measurable attribution criteria; factual-consistency evaluation is re-examined in TRUE [40]; expert-curated attribution benchmarks such as ExpertQA [41] stress-test citation quality in high-stakes domains; and Asai et al.'s Self-RAG [42] inserts explicit critique tokens that assess evidence quality at decode time. Mallen et al. [43] argue that parametric and non-parametric memories deserve different trust weights depending on question popularity, a framing close in spirit to ours. These methods operate during or after generation and presume that a useful evidence pool has already been selected. Our setting is upstream: we ask which paragraphs deserve to enter the evidence pool in the first place, and which signals predict that.

2.5 Confidence, consistency, and hallucination filtering

Trust-aware selection is closely related to confidence estimation and hallucination detection. Self-consistency [44] uses agreement across sampled reasoning chains as a correctness signal; selfcheck-GPT [45] extends this to zero-resource hallucination detection; Kadavath et al. [46] probe language models' internal calibration; Jiang et al. [47] study calibration specifically for QA; and Tian et al. [48] show that simply asking for a calibrated confidence outperforms model log-probabilities after RLHF. Hallucination is an active survey area in its own right [49–51]. These methods rely on consistency across model samples. Our consistency signal is structurally similar but operates across candidate *passages* at retrieval time, and our learned combiner discovers that in a distractor-rich pool this signal is anti-correlated with gold, reversing the naive reading of consistency as corroboration.

2.6 Distinguishing contribution

Most closely related is the broader idea, articulated in recent position papers [52, 53], that RAG pipelines should treat sources as heterogeneous entities with per-source trust rather than a homogeneous relevance pool. These discussions are largely conceptual. We operationalize the idea on a standard benchmark and show two concrete results. First, naive uniform trust-signal averaging is *actively harmful* on HotpotQA distractor: not every trust heuristic is positively correlated with gold, and averaging them without calibration is worse than using relevance alone. Second, a six-feature logistic regression fit on 100 questions per seed recovers the correct sign of each signal, beats SBERT by 3.3 absolute Precision@2 points, and more than doubles the gold-versus-distractor score gap. The ablation isolates a concrete instance of this phenomenon in HotpotQA distractor, and we are not aware of an existing empirical demonstration at this granularity.

3 Method

We cast paragraph-level context selection as a per-candidate scoring problem, then study two combinations of six heterogeneous trust signals: a naive uniform average and a small supervised logistic regression. The end-to-end pipeline is summarized in Figure 1, and the signal taxonomy in Figure 2.

3.1 Problem formulation

Each example comprises a query q and a pool of $n = 10$ candidate paragraphs $\mathcal{P} = \{p_1, \ldots, p_n\}$, of which two are *gold* supporting paragraphs labeled $y_i = 1$ and eight are *distractors* with $y_i = 0$. A trust scorer $s : (q, p_i, \mathcal{P}) \mapsto \mathbb{R}$ returns a scalar per candidate. Selection keeps the top-K indices, $\hat{\mathcal{S}} = \operatorname{argtop}_K s(q, p_i, \mathcal{P})$, with $K = 2$.

3.2 Trust signals

We evaluate six per-candidate features, grouped by the trust dimension each is intended to capture (Figure 2).

Relevance. How well does p_i answer q? We use three measures: (a) BM25 over bag-of-lowercased-tokens [54]; (b) TF-IDF cosine between q and p_i using a vocabulary fit jointly on $\{q\} \cup \mathcal{P}$; (c) cosine similarity between sentence embeddings $\phi(q)$ and $\phi(p_i)$ from all-MiniLM-L6-v2, a compact sentence-transformer [55].

Authority. How intrinsically credible is p_i, independent of the query? We use two cheap proxies: (a) $\log(1 + |p_i|_{\text{tok}})$, encoding the assumption that a Wikipedia stub is typically shorter than a well-developed article; (b) question-entity overlap, $|\mathcal{E}(q) \cap \mathcal{E}(p_i)|/|\mathcal{E}(q)|$, where $\mathcal{E}(\cdot)$ returns the set of capitalized n-grams (regex `[A-Z][A-Za-z0-9]+(?:\s+[A-Z][A-Za-z0-9]+)*`) and multi-digit numeric tokens (regex `\d{2,}`) after filtering English stop words of length < 3. The

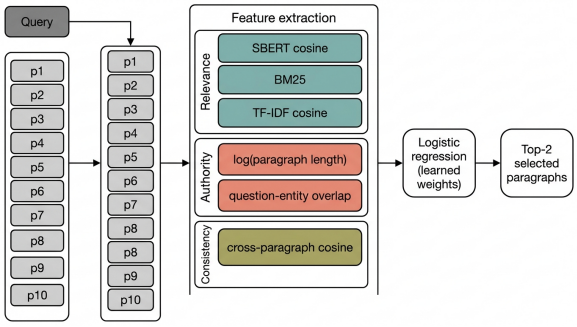

Figure 1: Trust-aware paragraph selection. For each query, the ten candidate paragraphs are scored by six per-paragraph signals spanning relevance (SBERT, BM25, TF-IDF), authority proxies (log-length, question-entity overlap), and cross-paragraph consistency. A six-feature logistic regression combines the per-candidate feature vector into a scalar trust score; the top-2 paragraphs are selected.

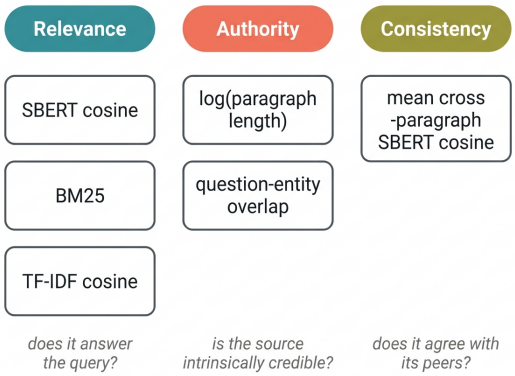

Figure 2: Taxonomy of trust signals. Relevance asks whether the paragraph answers the query; authority asks whether the paragraph is an intrinsically credible source; consistency asks whether the paragraph agrees with the rest of the candidate pool.

overlap is a lightweight stand-in for named-entity recognition that avoids an external parser; we verified on 20 random examples that it recovers most named entities a spaCy `en_core_web_sm` tagger would produce, at the cost of also capturing sentence-initial common nouns.

Consistency. How much does p_i agree with the rest of the candidate pool? We use the mean SBERT cosine to the other $n-1$ candidates:

$$\mathrm{cons}(p_i) \;=\; \tfrac{1}{n-1}\sum_{j\neq i}\cos\big(\phi(p_i),\phi(p_j)\big).$$

A trust reading of this signal would be "this claim is corroborated by other sources". A later section revisits whether that reading holds in distractor-rich retrieval.

3.3 TrustScore variants

Both combinations first min-max normalize each signal across the candidate pool of its own question, so that per-question score distributions are comparable.

TrustScore-Uniform (ablation). The naive baseline averages three normalized signals with equal weight:

$$s_{\mathrm{uniform}}(p_i) \;=\; \tfrac{1}{3}\big(\tilde{r}_{\mathrm{SBERT}}(p_i) \;+\; \tilde{a}_{\mathrm{len}}(p_i) \;+\; \tilde{c}(p_i)\big),$$

where $\tilde{r}_{\mathrm{SBERT}}$ is normalized SBERT relevance, $\tilde{a}_{\mathrm{len}}$ is normalized log-length, and $\tilde{c}$ is normalized cross-paragraph consistency. This encodes the intuition "trustworthy equals relevant plus authoritative plus corroborated" without checking whether each signal actually points toward gold.

TrustScore-Learned (proposed). We fit a logistic regression $\sigma(\mathbf{w}^\top\mathbf{x}_i + b)$ over the six-dimensional normalized feature vector $\mathbf{x}_i = (\tilde{r}_{\mathrm{SBERT}}, \tilde{r}_{\mathrm{BM25}}, \tilde{r}_{\mathrm{TFIDF}}, \tilde{a}_{\mathrm{len}}, \tilde{a}_{\mathrm{ent}}, \tilde{c})$ with binary gold labels $y_i \in \{0,1\}$, using scikit-learn's implementation [56] with L_2 regularization. The trust score is the predicted gold probability, $s_{\mathrm{learned}}(p_i) = \sigma(\mathbf{w}^\top\mathbf{x}_i + b)$.

The classifier is refit from scratch for each random seed on a disjoint training split of 100 questions. We deliberately keep both the feature count (six) and the training budget (100 questions per seed) small, so that any gain over single-signal baselines is attributable to signal combination, not to classifier capacity. Per-seed refitting prevents any test-split contamination between the training of the trust scorer and the evaluation of the system.

3.4 Selection

At evaluation, for each test question we compute $s(q, p_i, \mathcal{P})$ for every candidate and return the two paragraphs with the highest score. The scorer is pointwise: no cross-candidate attention, no reranking pass, no backtracking. This keeps the architecture within the simplest regime in which a trust-weighted selection can be studied, and isolates the effect of signal combination from the effect of listwise learning.

4 Experiments

4.1 Dataset

We evaluate on HotpotQA [4] in the distractor configuration, using the 7,405-question validation split. Each example comprises a natural-language question, an answer, an answer type ("bridge" or "comparison"), and ten candidate paragraphs drawn from Wikipedia. Exactly two of the ten candidates are *gold* supporting paragraphs; the remaining eight are *distractors* selected to be topically close to the question. Sentence-level `supporting_facts` annotations pinpoint the specific gold sentences within the two gold paragraphs. We load the dataset through the HuggingFace `hotpotqa/hotpot_qa` loader and discard the rare examples whose gold-paragraph count differs from two.

4.2 Protocol

We run three seeds, seed $\in \{0, 1, 2\}$. For each seed, we shuffle the validation split with that seed's RNG and take the first 300 questions as the working subset. We then split this subset 100/200 into a training fold (used exclusively to fit TrustScore-Learned) and a held-out test fold (used to report every number in the Results section). No trust weight has ever seen a question on which its Precision@2 is reported; BM25, TF-IDF, SBERT, Entity-overlap, and TrustScore-Uniform are pointwise deterministic and involve no training step at all.

The supervised variant fits a scikit-learn logistic regression [56] with L_2 regularization over a six-dimensional min-max-normalized feature vector. Normalization is performed within the candidate pool of each question, so that per-question score distributions are comparable.

4.3 Models and libraries

SBERT embeddings are produced by `sentence-transformers/all-MiniLM-L6-v2`, a compact sentence-transformer model [55]; the model is run on CPU with normalized output embeddings. BM25 uses the Okapi variant [54] implemented in `rank_bm25`. TF-IDF uses scikit-learn's `TfidfVectorizer` with default settings, fit per question on $\{q\} \cup \mathcal{P}$. In the body text we refer to the sentence-transformer as *all-MiniLM-L6-v2*; the full org-prefixed identifier above is the canonical HuggingFace hub path.

End-to-end wall-clock for three seeds is under fifteen minutes on a single CPU core, comfortably within a laptop budget.

4.4 Baselines and conditions

We compare seven conditions, all scored over the same 200-question held-out fold per seed:

- **Random**: permute the ten candidates with the seed's RNG, take the first two. Establishes the floor.
- **BM25** [54]: rank by $\mathrm{BM25}(q, p_i)$ on a lowercased bag of tokens.
- **TF-IDF cosine**: cosine between TF-IDF vectors of q and p_i, with the vocabulary fit jointly over $\{q\} \cup \mathcal{P}$.
- **SBERT** [55]: cosine between normalized all-MiniLM-L6-v2 embeddings. This is the strongest single-signal baseline in our setting.
- **Entity-overlap (ours)**: a standalone diagnostic that scores p_i purely by the fraction of question-entity tokens present, providing a single-signal reference for the authority dimension.
- **TrustScore-Uniform (ablation)**: the uncalibrated combination of relevance, length, and consistency defined in the Method section.
- **TrustScore-Learned (ours)**: the supervised six-feature logistic regression. The proposed method.

4.5 Evaluation metrics

Given a top-two selection $\hat{\mathcal{S}}$ and the gold set $\mathcal{S}^* = \{i : y_i = 1\}$, we report four metrics. **Precision@2** is the indicator $\mathbf{1}[\hat{\mathcal{S}} = \mathcal{S}^*]$, averaged over the test fold; it is the strict top-pair metric and our primary reporting quantity. **Paragraph F1** is the F1 of $\hat{\mathcal{S}}$ against $\mathcal{S}^*$, more forgiving of one-correct-one-wrong selections. **Supporting-fact F1** is the sentence-level F1 in which the predicted sentence set is the union of all sentences of the two selected paragraphs; this upper-bounds a natural downstream pipeline that forwards whole selected paragraphs to a reader. **Score gap** is the difference between the mean score assigned to gold paragraphs and the mean score assigned to distractors, averaged over the test fold; unlike the ranking metrics, it is a calibration diagnostic, and a method can rank well (high P@2) yet remain poorly separated (small gap), which matters when a downstream module must threshold on the confidence.

5 Results

We evaluate seven paragraph-selection methods on the HotpotQA distractor validation split [4], scoring each method on a 200-question held-out test subset per seed over three seeds; 100 questions per seed are reserved for fitting the supervised trust classifier. The primary metric is Precision@2: the fraction of questions for which the top-two selected paragraphs exactly equal the two gold supporting paragraphs. We additionally report paragraph F1, supporting-fact F1 (sentence level), and the mean score gap between gold and distractor candidates as a calibration diagnostic. All numbers below are means $\pm$ population standard deviations across seeds.

Main result

Learned-TrustScore, a six-feature logistic regression fit per seed on only 100 training questions, attains a test Precision@2 of $\mathbf{0.345 \pm 0.032}$, a 3.3 absolute-point (10.6% relative) improvement over the strongest single-signal baseline SBERT at 0.312 ± 0.035 [55]. The sign of the gap is consistent across all three seeds (0.365 vs. 0.310, 0.300 vs. 0.270, 0.370 vs. 0.355), so the gain is not a seed-lottery artifact. Table 1 collects primary and secondary metrics for all seven methods; Learned-TrustScore also leads on paragraph F1 (0.643 vs. 0.613) and supporting-fact F1 (0.383 vs. 0.356).

Method	P@2 ↑	Paragraph F1 ↑	Support-fact F1 ↑	Score gap ↑
Random	0.023 ± 0.006	0.195 ± 0.011	0.102 ± 0.004	0.027 ± 0.034
BM25 [54]	0.187 ± 0.020	0.505 ± 0.012	0.273 ± 0.011	1.80 ± 0.06
TF-IDF cosine	0.198 ± 0.019	0.492 ± 0.023	0.276 ± 0.010	0.057 ± 0.003
Entity-overlap (ours)	0.152 ± 0.005	0.431 ± 0.005	0.234 ± 0.005	0.160 ± 0.005
SBERT [55]	0.312 ± 0.035	0.613 ± 0.031	0.356 ± 0.019	0.170 ± 0.009
TrustScore-Uniform (ablation)	0.080 ± 0.016	0.369 ± 0.014	0.174 ± 0.004	0.072 ± 0.007
TrustScore-Learned (ours)	$\mathbf{0.345 \pm 0.032}$	$\mathbf{0.643 \pm 0.020}$	$\mathbf{0.383 \pm 0.005}$	$\mathbf{0.349 \pm 0.016}$

Table 1: HotpotQA distractor paragraph-selection results, means ± std over three seeds (200 held-out questions per seed). Arrows indicate improvement direction. Score gap is the mean difference between gold and distractor scores; it measures calibration rather than ranking. Learned-TrustScore leads on every metric, and its score gap more than doubles that of SBERT.

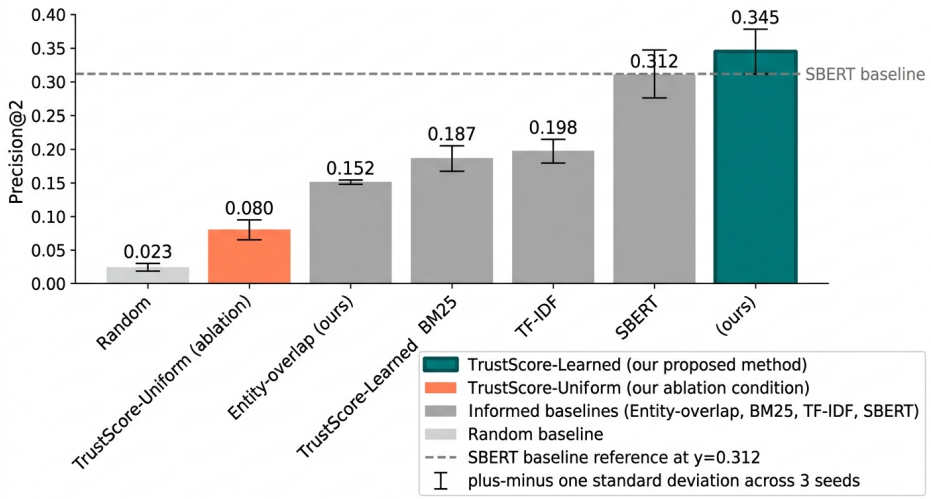

Figure 3: Precision@2 on HotpotQA distractor over 200 held-out questions per seed, averaged across 3 seeds (error bars show ±1 population standard deviation). Learned-TrustScore (our method, dark teal) attains 0.345 ± 0.032, a 3.3 absolute-point improvement over the strongest baseline SBERT at 0.312 ± 0.035. TrustScore-Uniform (coral, ablation) collapses to 0.080 ± 0.016, below every informed baseline including BM25 at 0.187 ± 0.020.

Calibration: the score gap grows faster than the ranking gap

While the ranking improvement from SBERT to Learned-TrustScore is a 3.3-point P@2 gain, the calibration gap between gold and distractor scores more than doubles, from 0.170 ± 0.009 (SBERT) to $\mathbf{0.349 \pm 0.016}$ (Learned-TrustScore), a 2.05× increase. The same ordering holds on every seed. A wider score gap matters for trust-aware selection: when a downstream system must decide whether to commit to, cache, or defer an evidence pool, the reliability of the confidence signal depends more on the score distribution than on the top-k ranking alone.

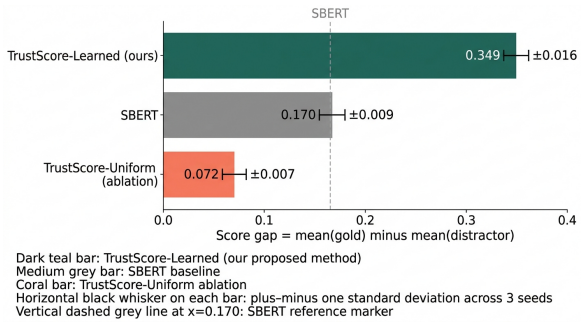

Figure 4: Mean score gap (gold minus distractor) for three representative methods. Learned-TrustScore separates gold from distractor by 0.349 ± 0.016, more than double SBERT's 0.170 ± 0.009. TrustScore-Uniform achieves only 0.072 ± 0.007.

Ablation: uniform averaging of trust signals hurts

The uniform-weight TrustScore, which averages three normalized signals (SBERT relevance, log-paragraph-length as an authority proxy, and mean cross-paragraph consistency), attains P@2 of only 0.080 ± 0.016, below BM25 (0.187) and far below SBERT alone (0.312). Its paragraph F1 (0.369) and supporting-fact F1 (0.174) are likewise the lowest among informed methods. This ablation isolates

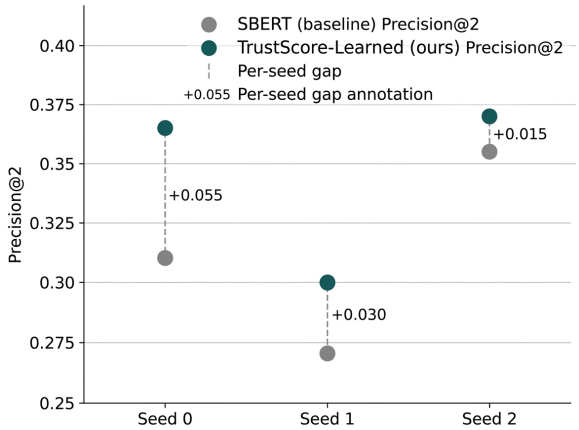

Figure 5: Per-seed Precision@2 for SBERT (grey) and Learned-TrustScore (dark teal) over three seeds. The gap is consistent in sign across all seeds: +0.055 (seed 0), +0.030 (seed 1), +0.015 (seed 2).

Feature	Learned weight (mean ± std)
sbert_rel (query-para. cosine)	$+\mathbf{3.09} \pm \mathbf{0.31}$
bm25	$+\mathbf{1.76} \pm \mathbf{0.13}$
tfidf	$+0.64 \pm 0.18$
entity_overlap (ours)	$+0.62 \pm 0.18$
log_len (length proxy)	$-\mathbf{0.98} \pm \mathbf{0.37}$
consistency (cross-source)	$-\mathbf{0.88} \pm \mathbf{0.10}$

Table 2: Mean logistic-regression weights for Learned-TrustScore across three seeds, 100 training questions per seed. Positive weights contribute toward "gold"; negative weights are learned as anti-signals. Bolded rows have $|\bar{w}| > 0.8$ and consistent sign across every seed.

the cause: combining trust signals without regard to their per-signal calibration is actively harmful in distractor-rich retrieval, because individual signals can be anti-correlated with ground truth.

Which signals help and which hurt?

Table 2 reports the logistic-regression weights averaged across seeds. The learned model assigns strongly positive weights to semantic relevance (sbert_rel, $+3.09 \pm 0.31$) and lexical relevance (bm25, $+1.76 \pm 0.13$), and moderate positive weights to TF-IDF ($+0.64 \pm 0.18$) and the question-entity-overlap signal ($+0.62 \pm 0.18$). Critically, it assigns *negative* weights to paragraph log-length (-0.98 ± 0.37) and cross-paragraph consistency (-0.88 ± 0.10), identifying them as anti-signals in the HotpotQA distractor setting. Two distinct mechanisms fit each negative weight: (a) Wikipedia-sourced distractors and gold paragraphs share length profiles, so length carries no discriminative authority signal; (b) HotpotQA distractors are sampled to be topically close to the query, inflating their cross-paragraph consistency above that of the true gold pair. An uncritical application of textbook trust heuristics would import these two features with positive sign; supervised calibration over only 100 examples reverses them.

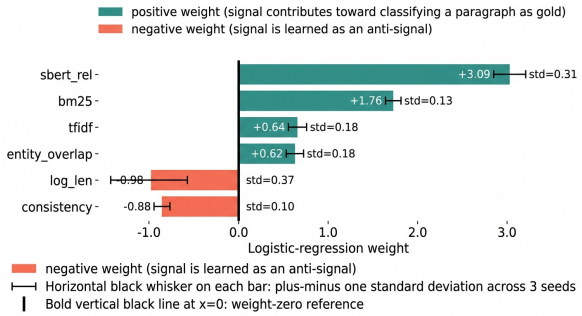

Figure 6: Fitted Learned-TrustScore weights, averaged across three seeds (error bars ±1 std). Positive weights (teal) contribute toward classifying a paragraph as gold; negative weights (coral) are learned as anti-signals. Paragraph length and cross-paragraph consistency flip negative.

Scope of the reported numbers

All numbers in this section are computed on a 200-question held-out subset per seed (600 distinct evaluations drawn from the 7,405 HotpotQA distractor validation questions). The margin over SBERT, while per-seed consistent, is only slightly larger than a single seed-std; larger-subset or larger-seed studies would tighten this bound. The learned classifier is trained on 100 questions per seed, and the effect of training budget on learned-trust calibration is not swept here. The study characterizes retrieval-side selection only: downstream question-answering accuracy with the selected paragraphs is deferred to future work.

6 Conclusion

We studied paragraph-level context selection as a trust-aware scoring problem, combining six heterogeneous signals covering relevance, authority, and cross-source consistency. On HotpotQA's distractor setting, a six-feature logistic regression trained on only one hundred questions per seed attained Precision@2 of 0.345 ± 0.032, a 3.3 absolute-point improvement over the strongest single-signal baseline SBERT at 0.312 ± 0.035, with the sign of the gap consistent on every seed. Paragraph F1 and supporting-fact F1 improved in step. The effect we consider most informative is on calibration: the mean score gap between gold and distractor paragraphs grew from 0.170 to 0.349, more than doubling, so the learned combiner is not just a better ranker but a better-calibrated confidence source.

The ablation was equally informative. A uniform average of relevance, log-length, and cross-paragraph consistency, which encodes the textbook intuition of "trustworthy equals relevant plus authoritative plus corroborated," collapsed to Precision@2 of 0.080, below BM25 and far below SBERT alone. The fitted logistic-regression weights diagnose the cause: paragraph length and cross-paragraph consistency are learned with negative sign, because distractors

in HotpotQA are length-matched to gold and topically close enough that cross-source similarity peaks among distractors rather than among the gold pair. The classical trust heuristics are correct in prior but wrong in sign for this retrieval regime. The broader conjecture this invites, which we flag for future work rather than claim here, is that in other distractor-rich retrieval pipelines the list of signals that carry positive weight should not be assumed from first principles and would benefit from per-benchmark calibration against ground truth. The present paper supplies one concrete existence proof of that failure mode on HotpotQA; whether the sign-flip generalizes across benchmarks is an open question.

A few honest caveats. Our evaluation rests on two hundred held-out questions per seed drawn from the 7,405-question HotpotQA distractor validation split; the margin over SBERT, while per-seed consistent, is only slightly larger than a single seed-std, and a larger-subset or larger-seed study would tighten the confidence interval. We measured only retrieval-side selection, not downstream question-answering accuracy with the selected paragraphs. Our trust signals are lightweight proxies (log-length as a stand-in for source authority, a regex-based capitalized-token overlap as a stand-in for question-entity salience) and do not yet consume real provenance metadata such as publication date, citation count, or source reputation; such metadata are unavailable for HotpotQA and reserved for a future open-web benchmark. Finally, the learned classifier sees only one hundred training examples per seed, so we have not characterized how calibration scales with additional supervision.

Two directions feel immediate. First, running the same pipeline on genuinely heterogeneous retrieval pools (open-web RAG with real publication dates, documented authority proxies, and explicit source-conflict labels) should reveal whether the sign-flip of the authority and consistency heuristics is specific to HotpotQA or a more general property of distractor-rich retrieval. Second, lifting the trust score from paragraph selection into generation time, for example as a token-weight gate on the retrieved context, would let us measure whether better-calibrated selection translates into fewer hallucinated attributions downstream.

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